

	r	r
10	A small sphere is moving in a vertical circle at constant speed. The magnitude of the resultant force on the sphere	
	A	at the bottom of the circle is greater than the resultant force at the top.
	B	at the top of the circle is greater than the resultant force at the bottom.
	C	decreases as the sphere moves from the bottom to the top of the circle.
	D	is the same at the top of the loop as it is at the bottom of the circle.

Solution: D

Resultant force = Centripetal force, mv^2 / r

Since the mass, speed and radius remain unchanged throughout, the resultant force is unchanged in magnitude.